

Optimization & Loss function

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Questions and comments



- Good * 2
- 👍
- I think I don't get the concept of how squared derivative connects to reduction of the unnecessary updates in gradient? But other parts are good and clear.
- Pretty good
- I feel very sorry that some university students care about their grades to such an extreme extent that you have to make such a decision. I don't think the world is completely fair; instead, I believe fairness is always relative. I think you are really passionate about teaching this course, and I feel very lucky to have you as our lecturer. Thank you!
- Learned a lot
- I have a hard time following all those complicated symbols
- 善👍讲的好👌

Questions and comments



- Very clear on the optimizer part, I love it. For the Kaggle part, I think we should more focus on the write up when submitting to blackboard, while if we use a "Accuracy Zone" to do the grading? Other than the first 3 in the contest can get a bit higher score, other students who attained a certain accuracy like 85% get the same score, then the next level maybe is 80, 75% something like that. Will this still be a fair grading?
  - Similar to the banding idea, what about the boundary case? 50% banding + 50% ranking?
- I would like to confirm one thing regarding the Scribing assignment: should we base our scribing work on the course topic we registered for, or the actual lecture date? I registered for the topic Optimization for October 7th, but the lecture on October 7th mainly covered CNN++ content. So whether I need to complete the scribing for the actual October 7th CNN++ lecture, or for the Optimization topic I signed up for?
  - Let's stick to the actual content of your signed date
- Very good
- NA

Questions and comments



- Very concrete concept, which i think the images help me understand what is going on
- So important knowledge
- Optimization is interesting
- Goooooood 6767
- Useful!
- Interesting and useful
- Good
- I like slide 27, which have the definition of all of the equations, which makes me very clear when i want to refer them. May I kindly ask if the future slides also include it? Thank you!!

Questions and comments



- Good
- Could we have a slight change to some
- Already good

About these two lectures

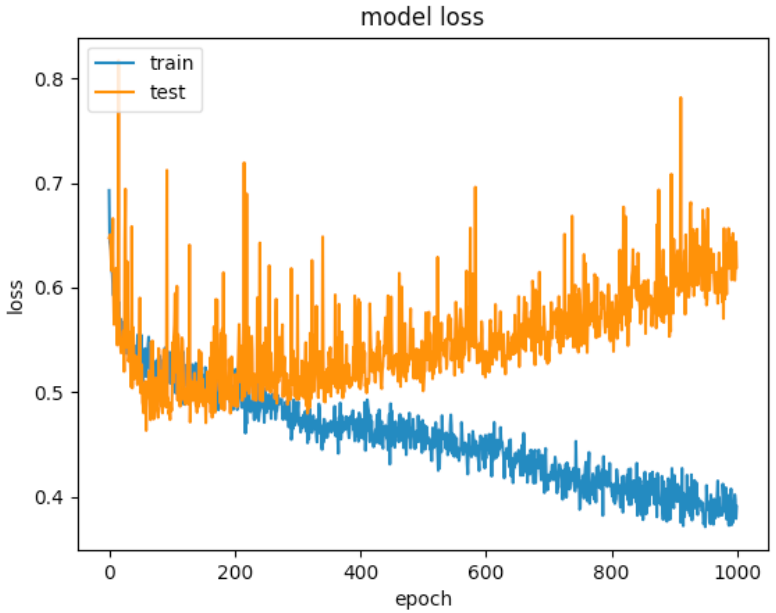


- Hope to give you more exposure to the details of the algorithm

- For practitioners, as long as you can understand the intuition behind them and know how to use them, it is fine enough
  - Math is a tool

Accumulate gradient

- The model is too large
- The batch is too small
  - Training can be very instable
- Accumulate the gradient for 10 iterations
  - For t = 1: 10, $\Delta w_{ij}^{(k)} \mathrel{+}= \alpha \frac{\partial Div(Y_{B_t}, d_{B_t})}{\partial w_{ij}^{(k)}}$
  - $w_{ij}^{(k)} = w_{ij}^{(k)} - \Delta w_{ij}^{(k)}$

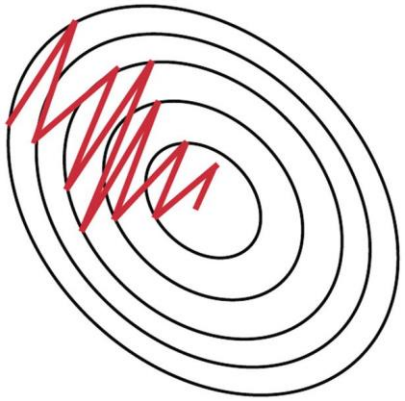


- Other solutions? Can we make the convergence faster and better?

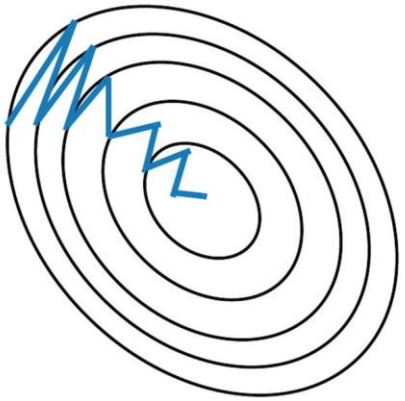
Momentum



- Reduce the oscillation
- Remember the update of the previous step
  - Step longer in directions with the same gradient sign
  - Step shorter in directions where the sign flips
- $w^t = w^{t-1} - \alpha \frac{\partial Loss}{\partial w^{t-1}}$
- $w^t = w^{t-1} - \alpha \frac{\partial Loss}{\partial w^{t-1}} + \beta \Delta w^{t-1}$
  - $\Delta w^{t-1} = w^{t-1} - w^{t-2}$
  - Typical $\alpha = 0.001$
  - Typical $\beta = 0.9$
- Allow a larger learning rate

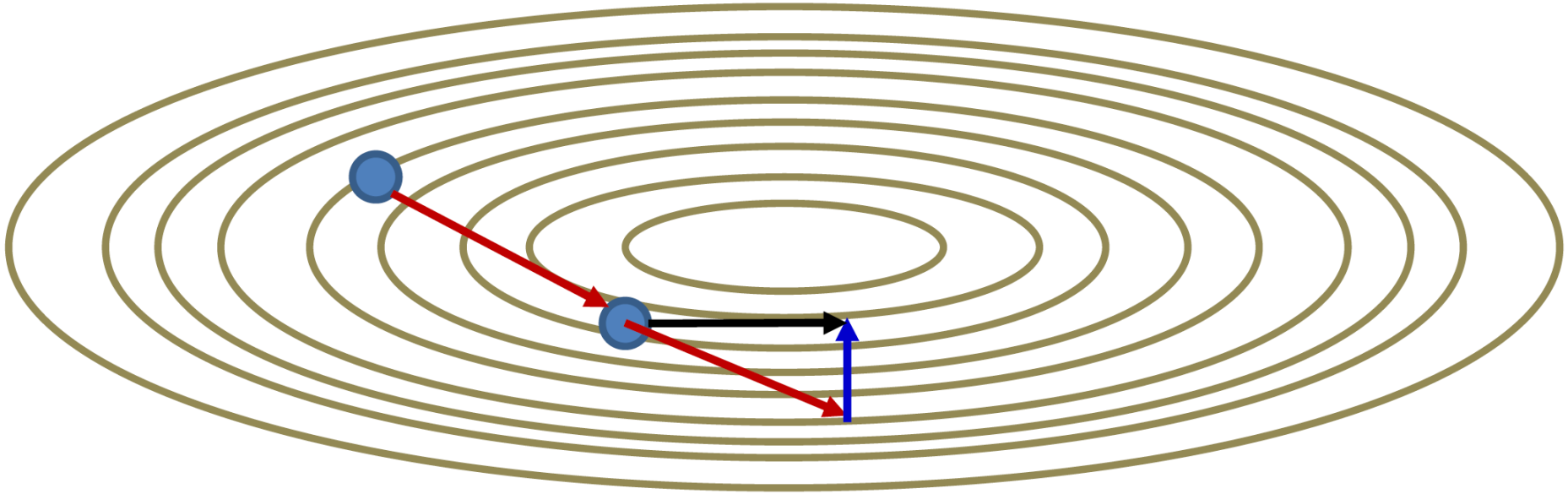


Stochastic Gradient Descent **without** Momentum



Stochastic Gradient Descent **with** Momentum

An alternative: Nestorov's accelerated gradient

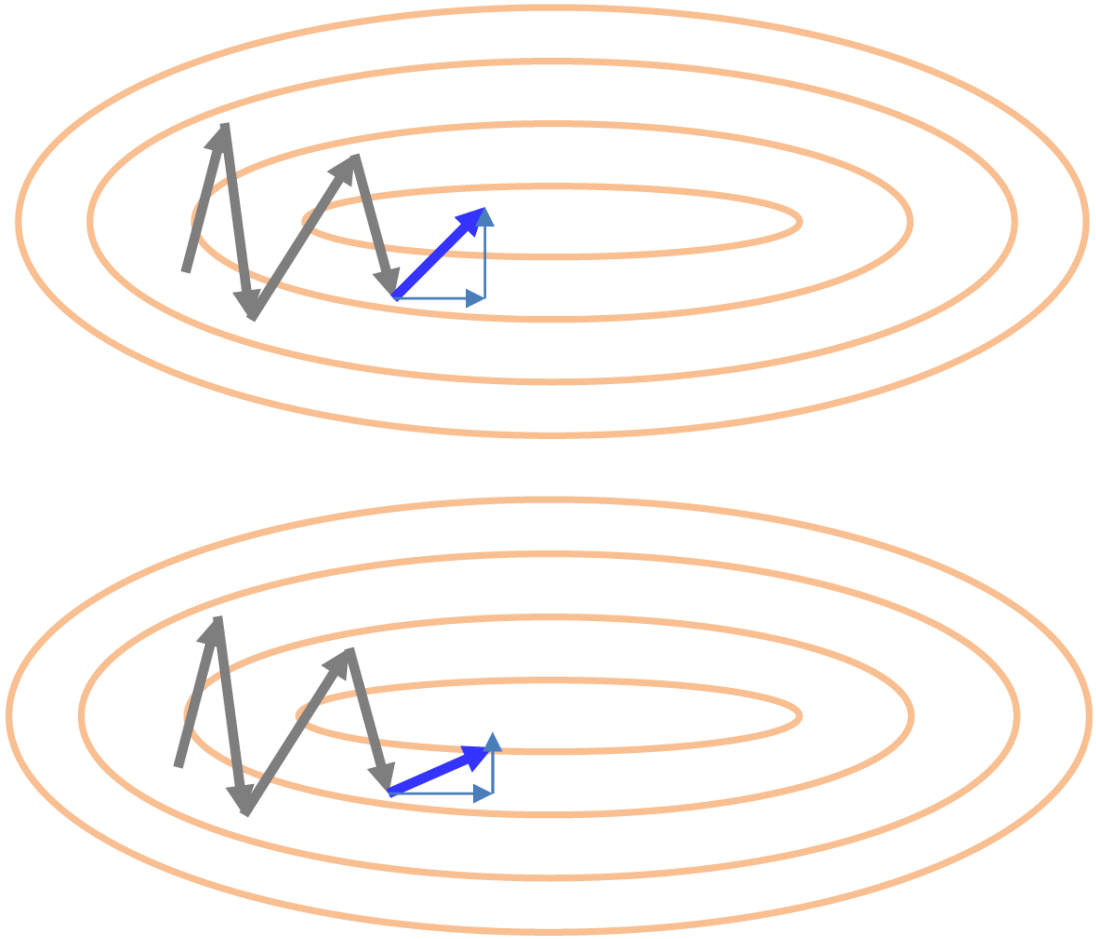


$$w^t = w^{t-1} - \alpha \frac{\partial Loss}{\partial (w^{t-1} + \beta \Delta w^{t-1})} + \beta \Delta w^{t-1}$$

For each iteration:

1. **First extend the previous step**
2. **Compute the gradient at the current location**
3. **Add the two together**

How to improve further?



Observation: Steps in the oscillatory direction show large total movement

Modify usual gradient-based update:
1. Scale updates in every component in inverse proportion to the total movement of that component in recent past
2. According to their unnormalized variation

RMS Prop



❖Root Mean Squared Propagation

❖$w^t = w^{t-1} - \frac{\alpha}{\sqrt{E[(\partial_w L)^2]_{t-1} + \epsilon}} \partial_{w^{t-1}} L$

- ➢$E[(\partial_w L)^2]_{t-1} = \gamma E[(\partial_w L)^2]_{t-2} + (1-\gamma)\left(\partial_{w^{t-1}} L\right)^2$
- ➢$E[(\partial_w L)^2]_0 = 0$
- ➢Typical $\gamma = 0.9$
- ➢Typical $\alpha = 0.001$
- ➢Typical $\epsilon = 10^{-7}$

❖The other steps are the same as the SGD, with backpropagation to calculate the gradients

RMS Prop



❖Root Mean Squared Propagation

How to improve it further?

❖$w^t = w^{t-1} - \frac{\alpha}{\sqrt{E[(\partial_w L)^2]_{t-1} + \epsilon}} \partial_{w^{t-1}} L$

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How to improve it further?

1. Is the running average a good estimation?
2. Did we forget the Momentum?

❖The other steps are the same as the SGD, with backpropagation to calculate the gradients

Put Momentum back



- $w^t = w^{t-1} - \alpha \frac{\partial Loss}{\partial w^{t-1}} + \beta \Delta w^{t-1}$ — Momentum
  - $\Delta w^{t-1} = w^{t-1} - w^{t-2}$
- $w^t = w^{t-1} - \frac{\alpha}{\sqrt{E[(\partial_w L)^2]_{t-1} + \epsilon}} \partial_{w^{t-1}} L$ — RMS Prop
  - $E[(\partial_w L)^2]_{t-1} = \gamma E[(\partial_w L)^2]_{t-2} + (1-\gamma)\left(\partial_{w^{t-1}} L\right)^2$
  - $E[(\partial_w L)^2]_0 = 0$
- **New formula**
- $w^t = w^{t-1} - \frac{\alpha}{\sqrt{E[(\partial_w L)^2]_{t-1} + \epsilon}} \partial_{w^{t-1}} L + \beta \Delta w^{t-1}$
  - $\Delta w^{t-1} = w^{t-1} - w^{t-2}$
  - $E[(\partial_w L)^2]_{t-1} = \gamma E[(\partial_w L)^2]_{t-2} + (1-\gamma)\left(\partial_{w^{t-1}} L\right)^2$
  - $E[(\partial_w L)^2]_0 = 0$

Look at the Momentum again



- $w^t = w^{t-1} - \alpha \frac{\partial Loss}{\partial w^{t-1}} + \beta \Delta w^{t-1}$
  - $\Delta w^{t-1} = w^{t-1} - w^{t-2}$
- $w^t = w^{t-1} - \alpha \frac{\partial Loss}{\partial w^{t-1}} + \beta (w^{t-1} - w^{t-2})$
- $w^t = w^{t-1} - \alpha (\frac{\partial Loss}{\partial w^{t-1}} - \frac{\beta}{\alpha} (w^{t-1} - w^{t-2}))$
- $(w^{t-1} - w^{t-2}) = -\alpha \frac{\partial Loss}{\partial w^{t-2}} + \beta (w^{t-2} - w^{t-3})$
- $w^t = w^{t-1} - \alpha (\frac{\partial Loss}{\partial w^{t-1}} - \frac{\beta}{\alpha} (-\alpha \frac{\partial Loss}{\partial w^{t-2}} + \beta (w^{t-2} - w^{t-3})))$
- $w^t = w^{t-1} - \alpha (\frac{\partial Loss}{\partial w^{t-1}} + \beta \frac{\partial Loss}{\partial w^{t-2}} - \frac{\beta^2}{\alpha} (w^{t-2} - w^{t-3}))$
- $w^t = w^{t-1} - \alpha (\frac{\partial Loss}{\partial w^{t-1}} + \beta \frac{\partial Loss}{\partial w^{t-2}} + \cdots + \beta^{t-1} \frac{\partial Loss}{\partial w^0} + 0)$

Look at the Momentum again



- $w^t = w^{t-1} - \alpha \frac{\partial Loss}{\partial w^{t-1}} + \beta \Delta w^{t-1}$
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A running average of derivative

We can unify the expression

Unify the expression



- $w^t = w^{t-1} - \frac{\alpha}{\sqrt{E[(\partial_w L)^2]_{t-1} + \epsilon}} E[\partial_w L]_{t-1}$
  - $E[(\partial_w L)^2]_{t-1} = \gamma E[(\partial_w L)^2]_{t-2} + (1-\gamma)\left(\partial_{w^{t-1}} L\right)^2$
  - $E[\partial_w L]_{t-1} = \delta E[\partial_w L]_{t-2} + (1-\delta)\partial_{w^{t-1}} L$
  - $E[(\partial_w L)^2]_0 = 0$
  - $E[\partial_w L]_0 = 0$

The problem of running average



- $w^t = w^{t-1} - \frac{\alpha}{\sqrt{E[(\partial_w L)^2]_{t-1}+\epsilon}} E[\partial_w L]_{t-1}$
  - $E[(\partial_w L)^2]_{t-1} = \gamma E[(\partial_w L)^2]_{t-2} + (1-\gamma)\left(\partial_{w^{t-1}} L\right)^2$
  - $E[\partial_w L]_{t-1} = \delta E[\partial_w L]_{t-2} + (1-\delta)\partial_{w^{t-1}} L$
  - $E[(\partial_w L)^2]_0 = 0$
  - $E[\partial_w L]_0 = 0$
- Is the running average a good estimation of the expectation?
- Denote the running average as m_t
- Denote the value of each observation as g_t
- $m_t = \delta m_{t-1} + (1-\delta) g_t = (1-\delta) g_t + (1-\delta)\delta g_{t-1} + \cdots + (1-\delta)\delta^{t-1} g_1$
- If running average is good, then $E[m_t] = E[g_t]$

The expectation of the running average



- If running average is good, then $E[m_t] = E[g_t]$
- $m_t = \delta m_{t-1} + (1-\delta) g_t = (1-\delta) g_t + (1-\delta)\delta g_{t-1} + \cdots + (1-\delta)\delta^{t-1} g_1$
- $m_t = (1-\delta) \sum_{i=0}^{t} \delta^{t-i} g_i$
- $E[m_t] = E\left[(1-\delta) \sum_{i=0}^{t} \delta^{t-i} g_i\right]$
- $E[m_t] = E\left[(1-\delta) \sum_{i=0}^{t} \delta^{t-i} (g_t + \varsigma)\right]$
- $E[m_t] = E[g_t](1-\delta) \sum_{i=0}^{t} \delta^{t-i} + \varsigma'$ Sum of a finite geometric series
- $E[m_t] = E[g_t](1-\delta^t) + \varsigma'$

The expectation of the running average



- If running average is good, then $E[m_t] = E[g_t]$
- $m_t = \delta m_{t-1} + (1-\delta) g_t = (1-\delta) g_t + (1-\delta)\delta g_{t-1} + \cdots + (1-\delta)\delta^{t-1} g_1$
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- $E[m_t] = E[g_t](1-\delta^t) + \varsigma'$

Running average correction



- $w^t = w^{t-1} - \frac{\alpha}{\sqrt{E[(\partial_w L)^2]_{t-1}+\epsilon}} E[\partial_w L]_{t-1}$
  - $E[(\partial_w L)^2]_{t-1} = \gamma E[(\partial_w L)^2]_{t-2} + (1-\gamma)\left(\partial_{w^{t-1}} L\right)^2$
  - $E[\partial_w L]_{t-1} = \delta E[\partial_w L]_{t-2} + (1-\delta)\partial_{w^{t-1}} L$
  - $E[(\partial_w L)^2]_0 = 0$
  - $E[\partial_w L]_0 = 0$
- $m_t = \delta m_{t-1} + (1-\delta) g_t = (1-\delta) g_t + (1-\delta)\delta g_{t-1} + \cdots + (1-\delta)\delta^{t-1} g_1$
- $E[m_t] = E[g_t](1-\delta^t) + \varsigma'$
- **The running average should be corrected** $\frac{E[(\partial_w L)^2]_{t-1}}{1-\gamma^{t-1}}, \frac{E[\partial_w L]_{t-1}}{1-\delta^{t-1}}$

When will the correction effect be the largest?

Adam: Adaptive moment estimation



- $w^t = w^{t-1} - \frac{\alpha}{\sqrt{\hat{E}[(\partial_w L)^2]_{t-1} + \epsilon}} \hat{E}[\partial_w L]_{t-1}$
  - $\hat{E}[(\partial_w L)^2]_{t-1} = \frac{E[(\partial_w L)^2]_{t-1}}{1-\gamma^{t-1}}$
  - $\hat{E}[\partial_w L]_{t-1} = \frac{E[\partial_w L]_{t-1}}{1-\delta^{t-1}}$
  - $E[(\partial_w L)^2]_{t-1} = \gamma E[(\partial_w L)^2]_{t-2} + (1-\gamma)\left(\partial_{w^{t-1}} L\right)^2$
  - $E[\partial_w L]_{t-1} = \delta E[\partial_w L]_{t-2} + (1-\delta)\partial_{w^{t-1}} L$
  - $E[(\partial_w L)^2]_0 = 0$
  - $E[\partial_w L]_0 = 0$
  - Typical $\alpha = 0.001$
  - Typical $\gamma = 0.999$
  - Typical $\delta = 0.9$
  - Typical $\epsilon = 10^{-7}$

First-order moment: mean
Second-order moment: variance

Can be considered as RMSprop with momentum

Currently, the most famous optimizer

Does not necessarily to be the best

Sometimes, SGD with carefully tunned learning rate can be also good

To make you awake



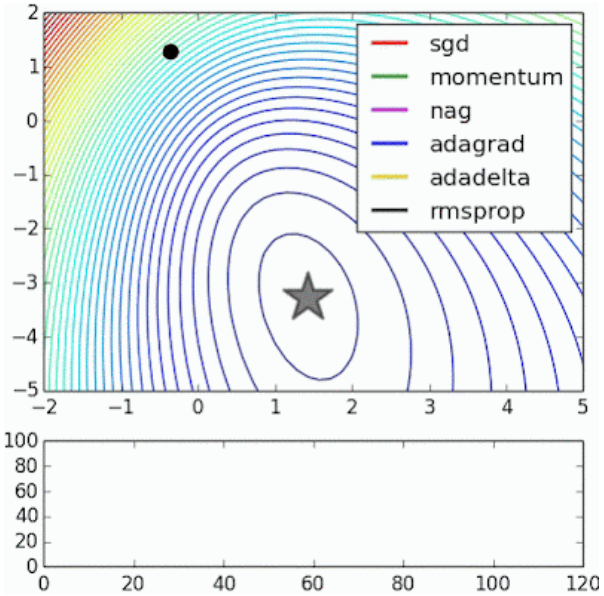
https://ureply.mobi/teacher

Other similar optimizers

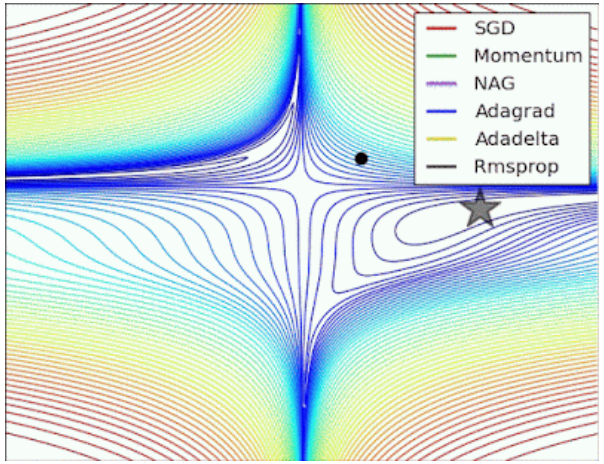


- Adagrad
- AdaDelta
- AdaMax
- …
- In practice, all roughly equivalent in performance

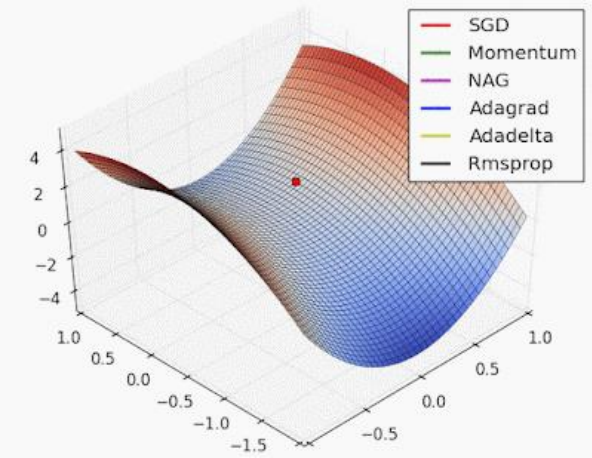
Visualization of different optimization methods



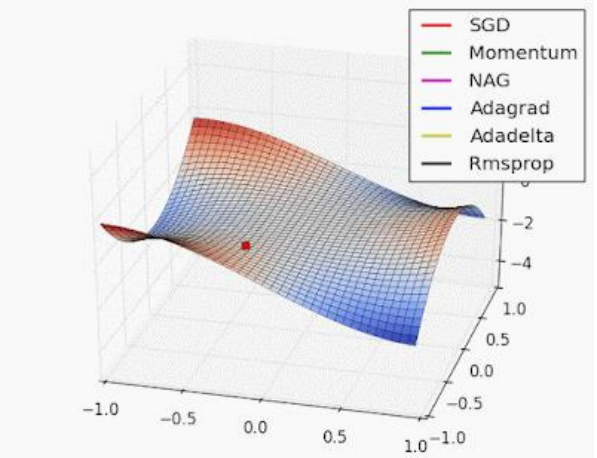
Noisy moons



Beale's function

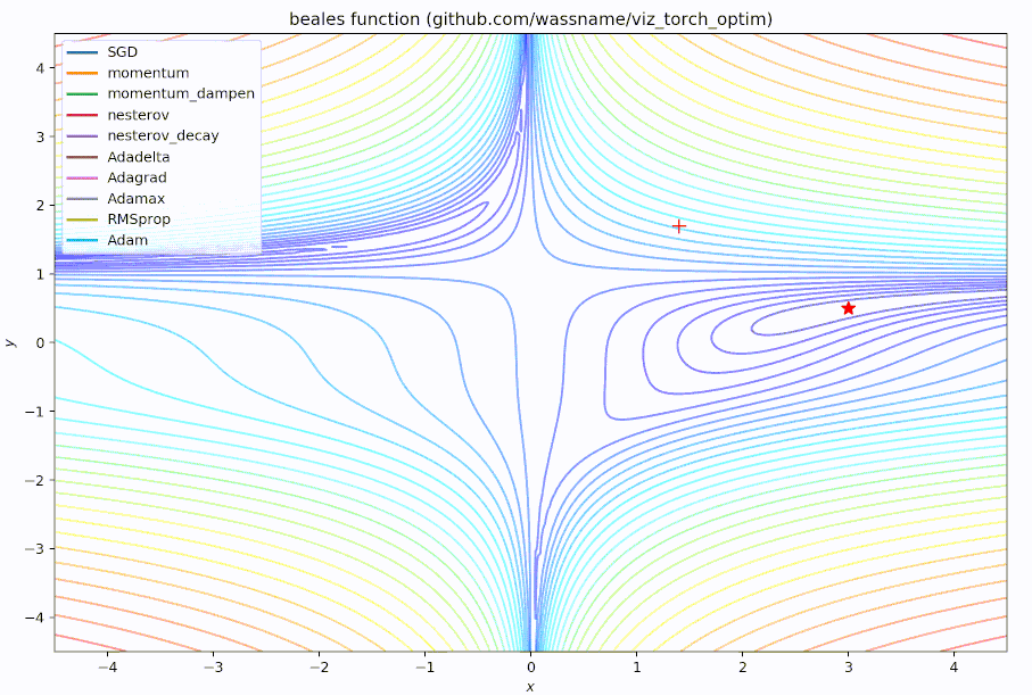


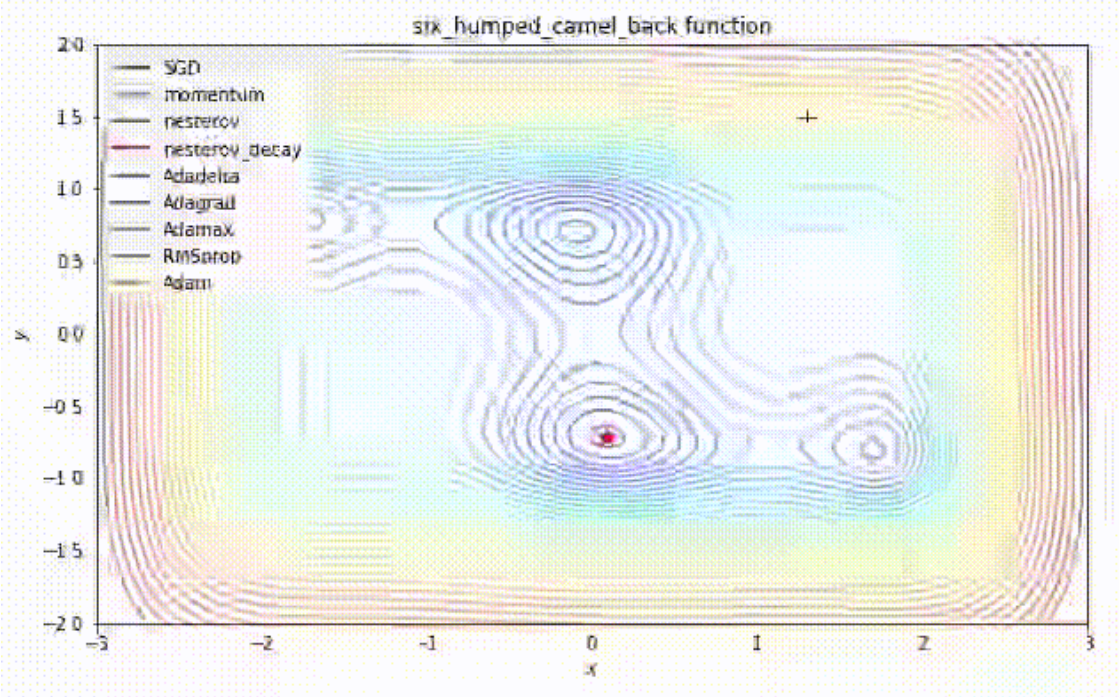
Long valley



Saddle point

Visualization of different optimization methods





Real-life is much more complicated than those scenarios

Optimizers in PyTorch



❖ https://pytorch.org/docs/stable/optim.html

Constructing it

To construct an `Optimizer` you have to give it an iterable containing the parameters (all should be `Variable` s) to optimize. Then, you can specify optimizer-specific options such as the learning rate, weight decay, etc.

Example:

```
optimizer = optim.SGD(model.parameters(), lr=0.01, momentum=0.9)
optimizer = optim.Adam([var1, var2], lr=0.0001)
```

Optimizers in PyTorch



- https://pytorch.org/docs/stable/generated/torch.optim.Adam.html#torch.optim.Adam

ADAM

CLASS torch.optim.Adam(*params*, *lr=0.001*, *betas=(0.9, 0.999)*, *eps=1e-08*, *weight_decay=0*, *amsgrad=False*, ***, *foreach=None*, *maximize=False*, *capturable=False*, *differentiable=False*, *fused=False*) [SOURCE]

Implements Adam algorithm.

Today's agenda



- Learning rate scheduling

- Applications & Loss functions

Learning rate scheduling



- All the popular methods root from GD
  - They tune the learning rate ratio and directions
  - $w^t = w^{t-1} - \frac{\alpha}{\sqrt{\hat{E}[(\partial_w L)^2]_{t-1} + \epsilon}} \hat{E}[\partial_w L]_{t-1}$

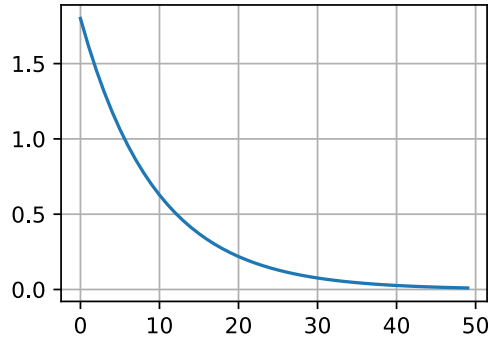
- Why not tune the learning rate by ourselves?

- At the beginning: large learning rate, increase speed

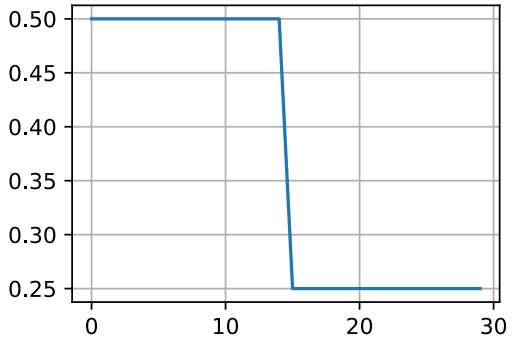
- Towards the end: small learning rate, improve accuracy

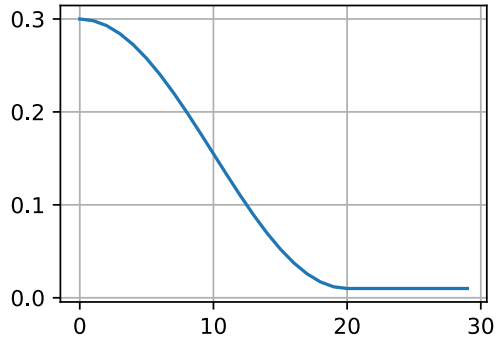
Learning rate scheduling

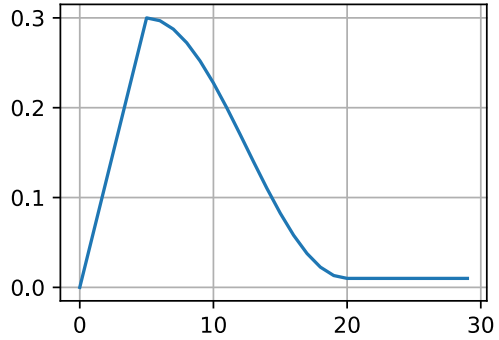
- In the old days, we set the learning rate manually
  - We reduce the learning rate by a factor of 10 every XX epochs



- Learning rate decay scheduler interface
  - Factor scheduler
    - $\alpha_{t+1} = \max(\alpha_{min}, \alpha_t * \tau)$
  - Multi factor scheduler
    - $\alpha_{t+1} = \alpha * \tau$, every several epochs
  - Cosine scheduler
    - $\alpha_t = \alpha_T + \frac{\alpha_0 - \alpha_T}{2}(1 + \cos(\frac{\pi t}{T}))$
    - α_0: initial rate, α_T: target rate
  - Warmup
    - Unstable optimization problem, small learning rate first







Learning rate scheduling in PyTorch



❖ https://pytorch.org/docs/stable/optim.html

How to adjust learning rate

`torch.optim.lr_scheduler` provides several methods to adjust the learning rate based on the number of epochs. `torch.optim.lr_scheduler.ReduceLROnPlateau` allows dynamic learning rate reducing based on some validation measurements.

Learning rate scheduling should be applied after optimizer's update; e.g., you should write your code this way:

Example:

```
model = [Parameter(torch.randn(2, 2, requires_grad=True))]
optimizer = SGD(model, 0.1)
scheduler = ExponentialLR(optimizer, gamma=0.9)

for epoch in range(20):
    for input, target in dataset:
        optimizer.zero_grad()
        output = model(input)
        loss = loss_fn(output, target)
        loss.backward()
        optimizer.step()
    scheduler.step()
```

To make you awake

https://ureply.mobi/teacher

Today's agenda



- Learning rate scheduling

- Applications & Loss functions

How to develop a GOOD machine learning method?



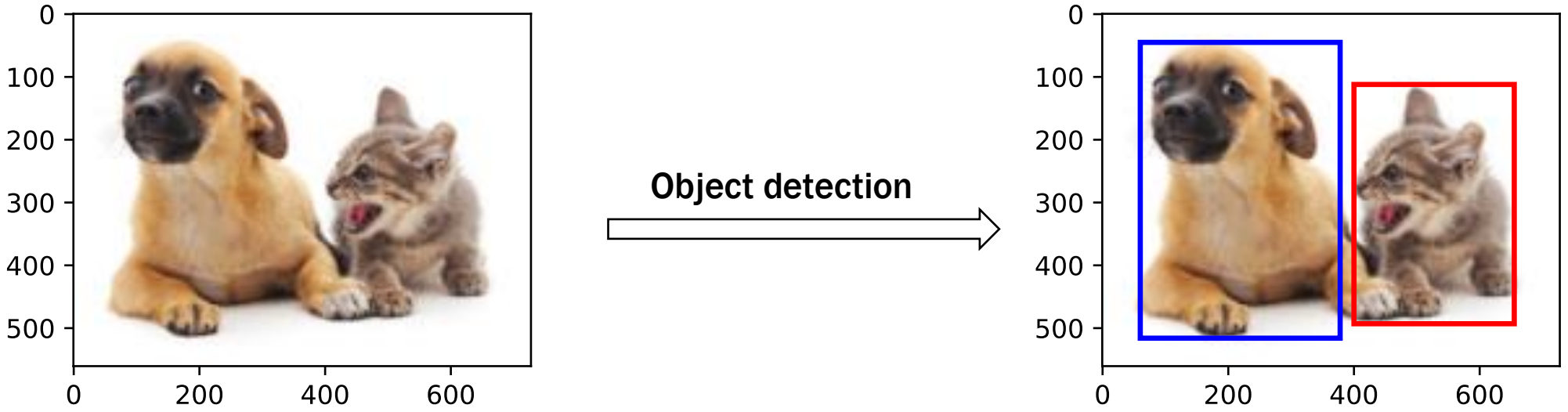
- Data
- Computational machine
- A suitable model
- Well-defined loss function
- Optimization
- Generalization
- Evaluation
- Deployment

Given a topic or problem:

1. Check whether you have the data, no matter annotated or unannotated, and machine, not matter good or bad
2. Design the model and loss function. The loss function should be related to the goal of your problem
3. Overfit the model
4. Other tricks and techniques to improve the method

Not just classification

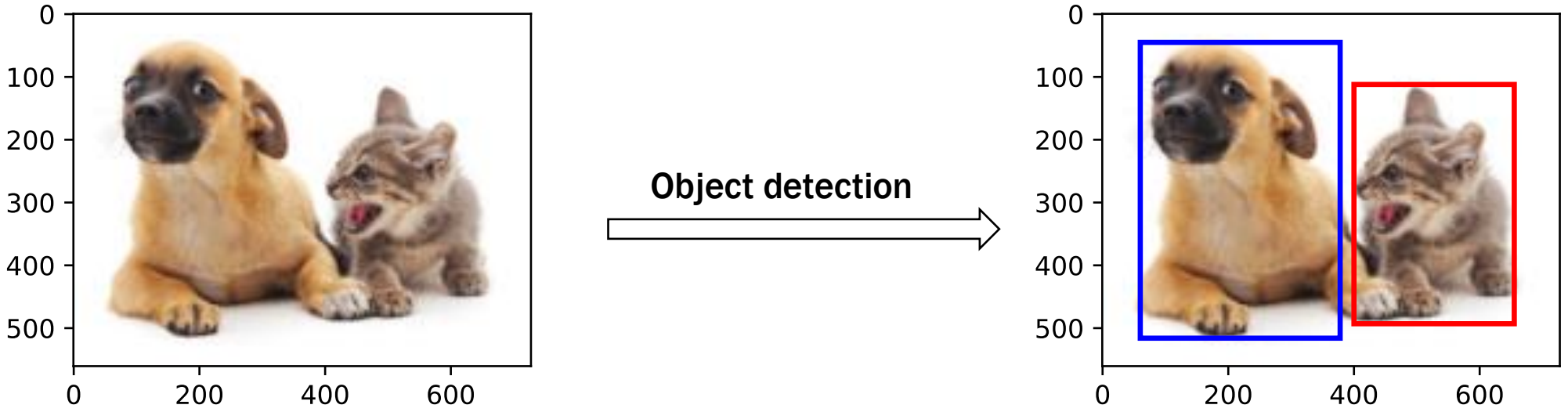
Examples of designing deep learning methods



1. Recognize the object of interest
2. Report their positions

Object detection

1. Recognize the object of interest
2. Report their positions

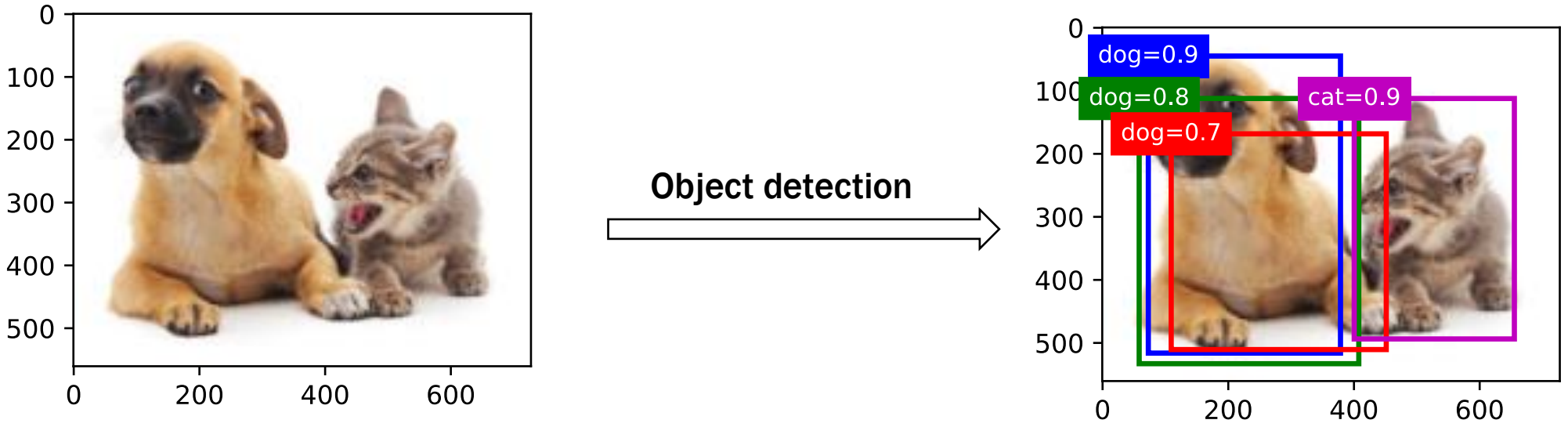


Think of the inference part first. Give an image:

1. Our method will generate lots of boxes
2. For each box, we assign a label and a confidence score
3. For each object, we use the box with the highest confidence score

Object detection

1. Recognize the object of interest
2. Report their positions

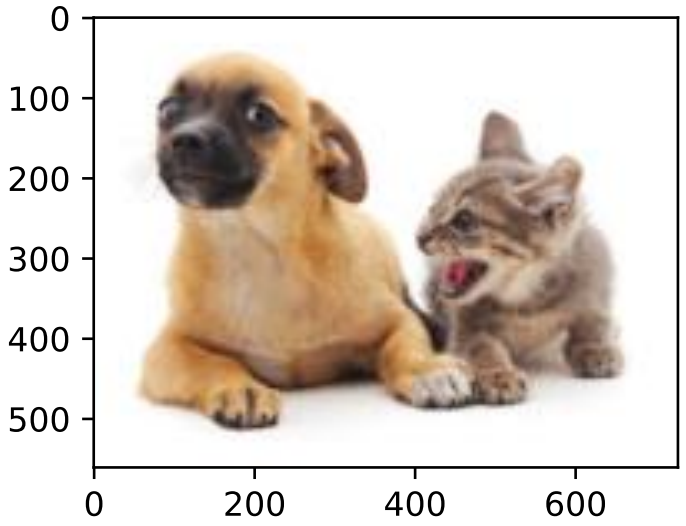


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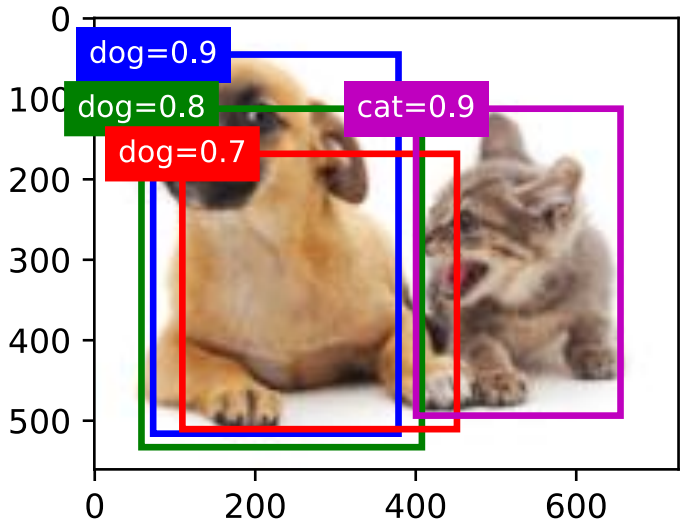
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Object detection

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Object detection

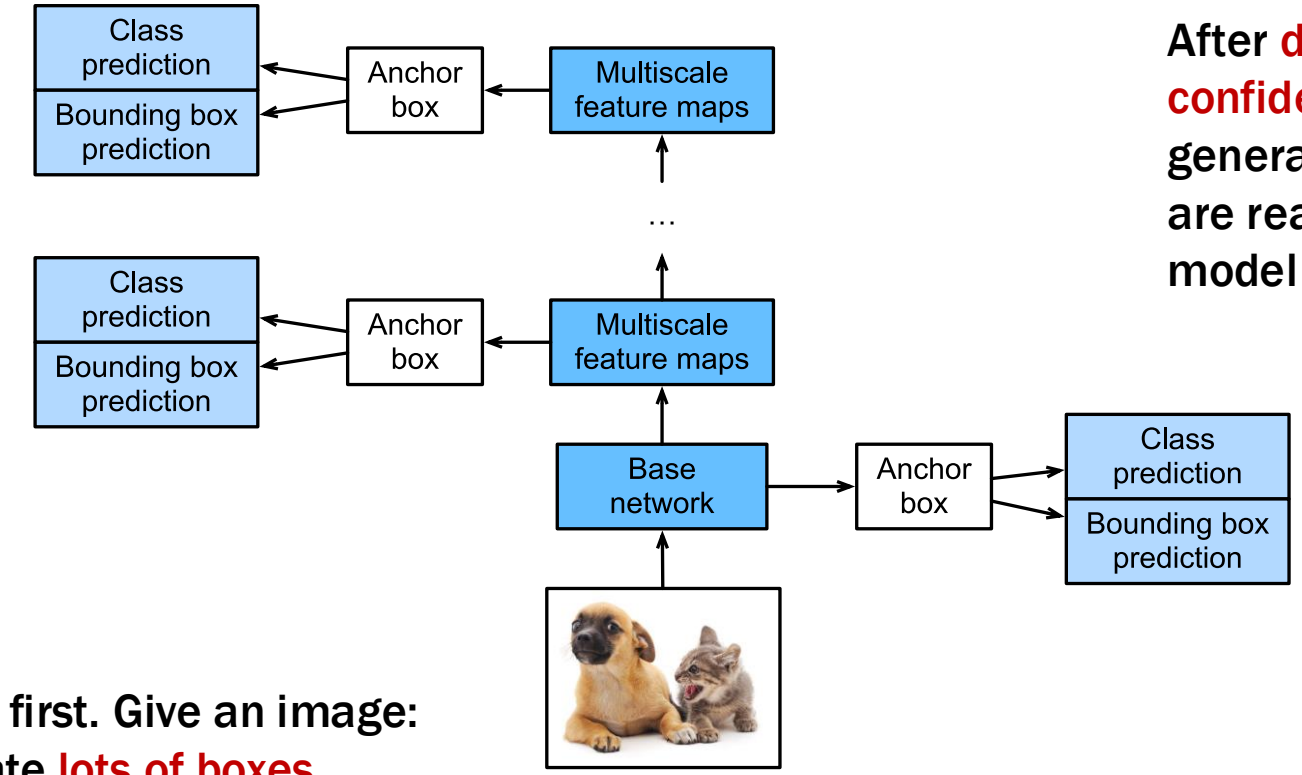


Think of the inference part first. Give an image:

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Loss function: classification and confidence score regression

Object detection



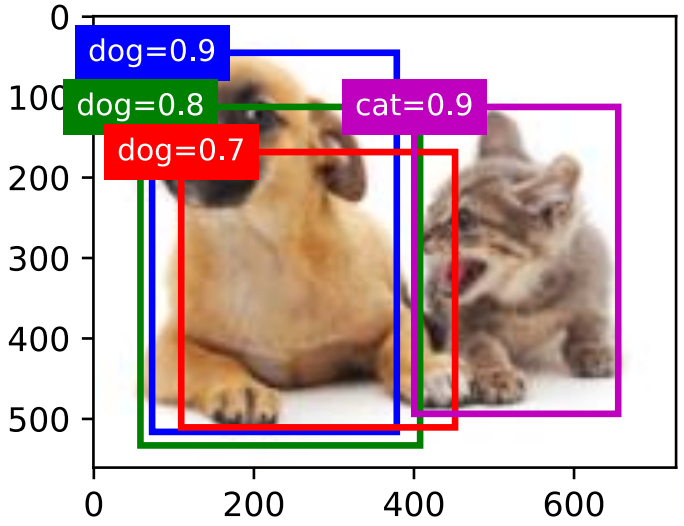
After defining the confidence score, and generate the data, we are ready to train the model

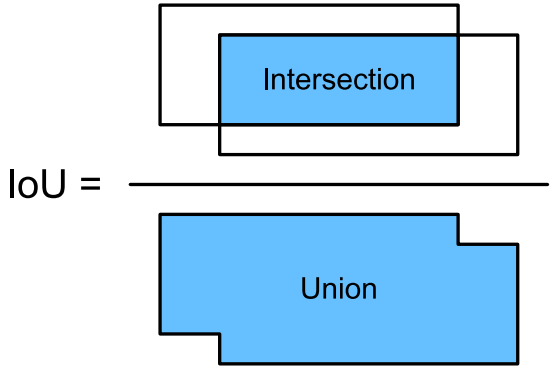
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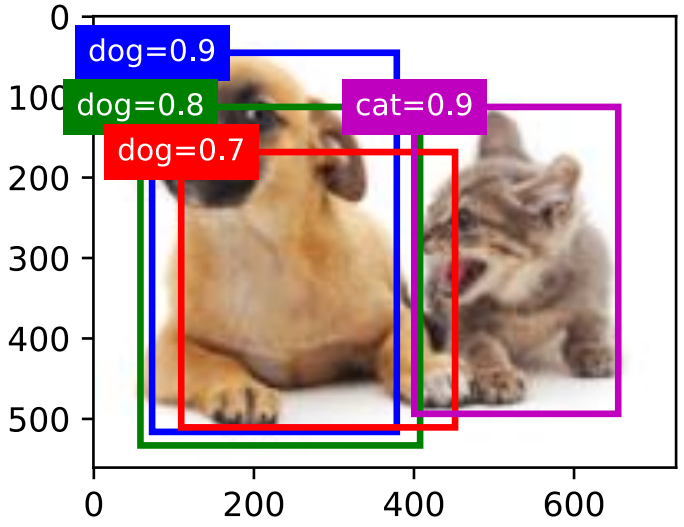
Loss function: classification and confidence score regression

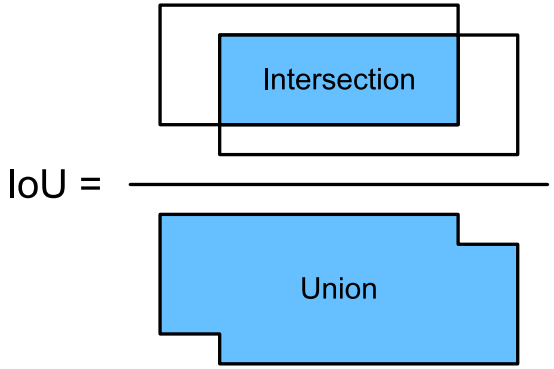
Define confidence score





Define confidence score





People will regress over the offset

$$\left(\frac{\frac{x_b - x_a}{w_a} - \mu_x}{\sigma_x}, \frac{\frac{y_b - y_a}{h_a} - \mu_y}{\sigma_y}, \frac{\log \frac{w_b}{w_a} - \mu_w}{\sigma_w}, \frac{\log \frac{h_b}{h_a} - \mu_h}{\sigma_h} \right)$$

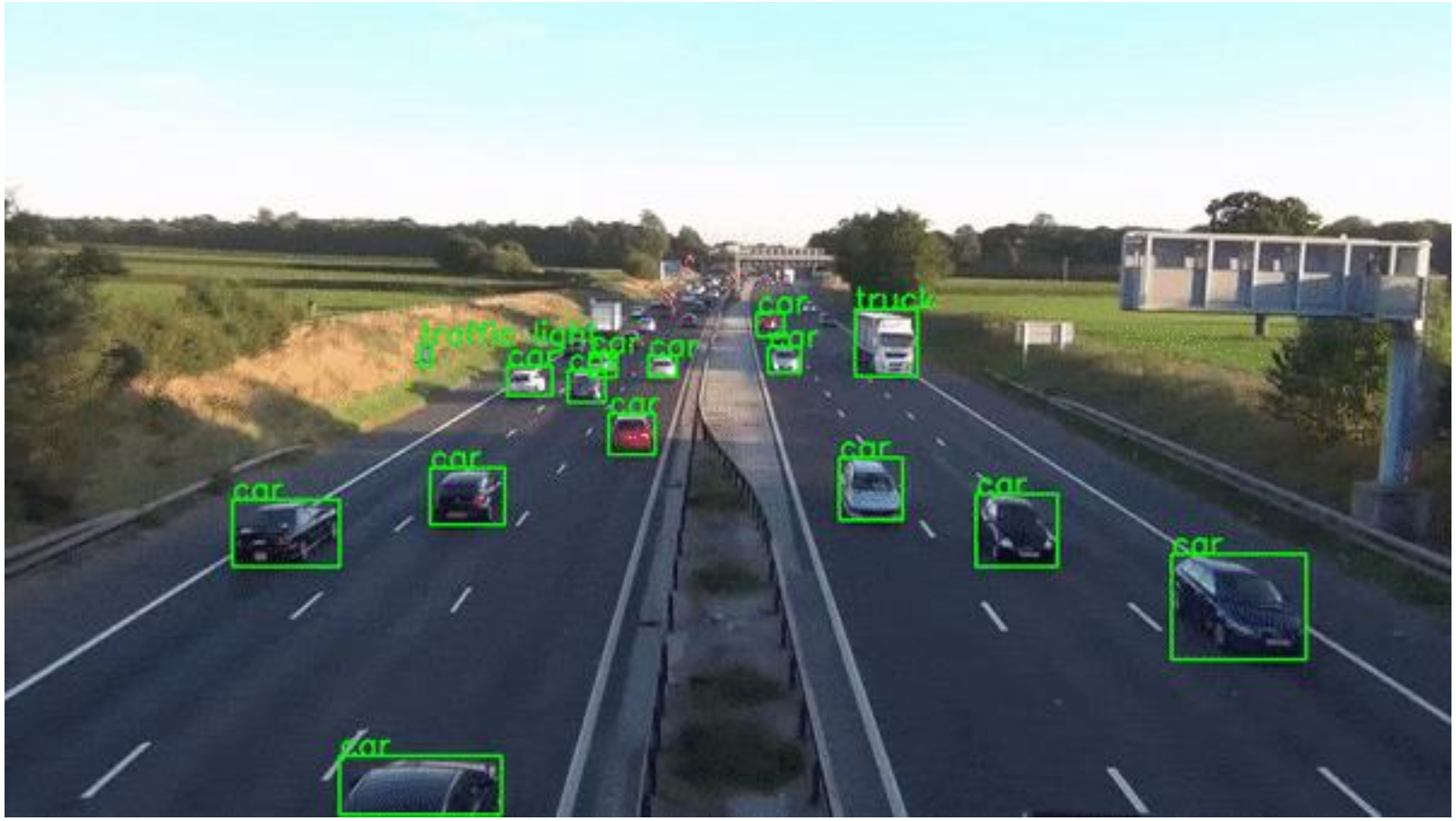
Leftover problems



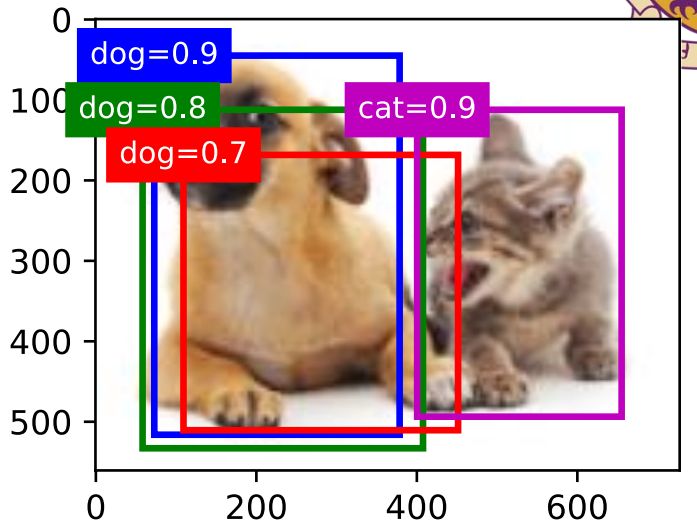
- ??

Leftover problems

❖??



Leftover problems

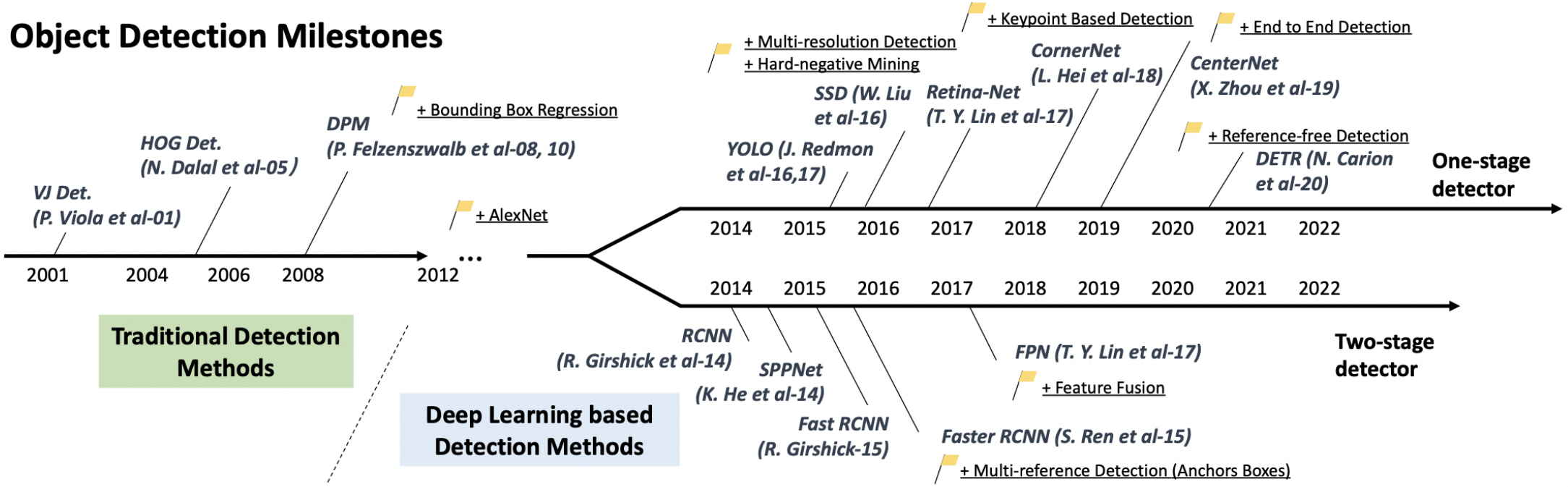


- How many box candidates should be generated?
- How to sample boxes more efficiently?
- Do we need to run the deep learning model for each box?
- How to balance accuracy and efficiency?
- How to incorporate the pixel-level information, if we have it?

To make you awake

https://ureply.mobi/teacher

Object detection

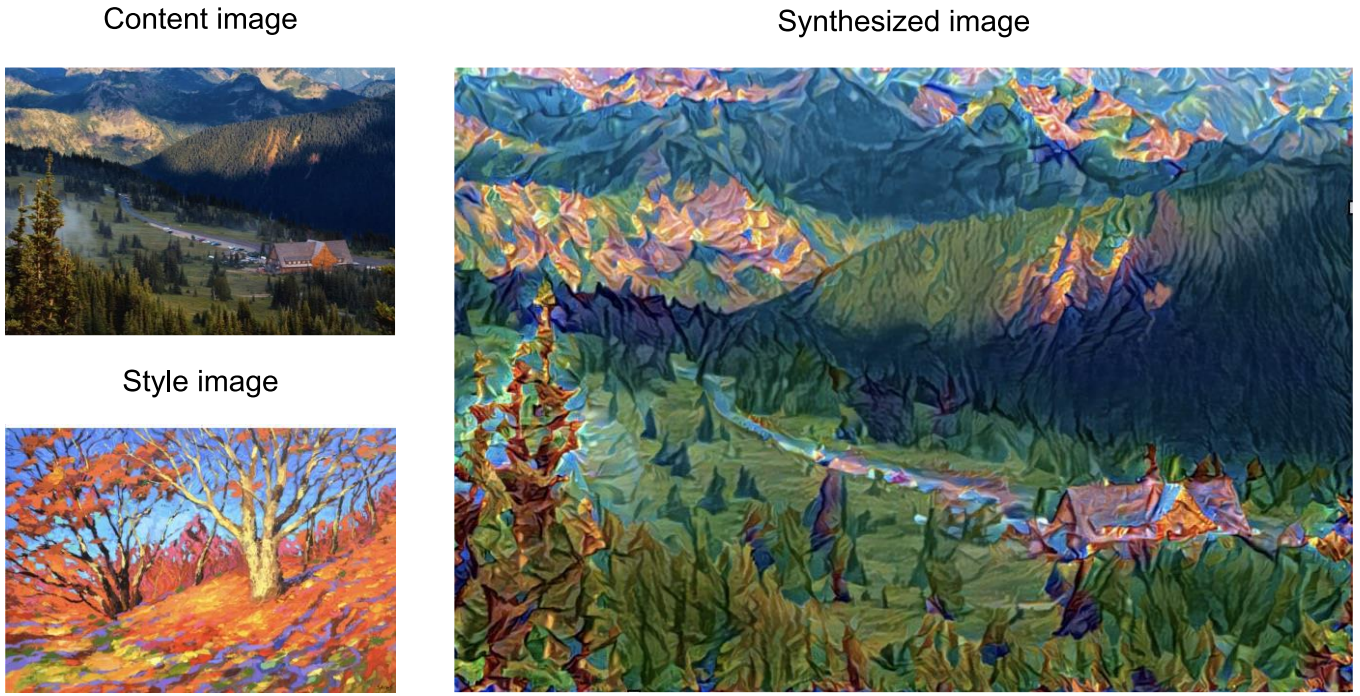


Four main tasks in computer vision



- Image classification
- Object detection
- Segmentation
- Image generation
- Other variants
  - Face recognition, Scene reconstruction, Edge detection etc.

Examples of designing deep learning methods



Style transfer

Style transfer

Input

Content image



Style image



Synthesized image



Output

Style: kind of feature map in the deep learning model. So, we need a deep learning model

Style transfer

Input

Content image



Style image



Synthesized image



Output

Feature map in the deep learning model

The image is similar to both input images. We should reduce their difference in the extract feature maps. Of course, on different layers for the content image and style image

Style transfer

Input

Content image



Style image



Synthesized image



Output

Feature map in the deep learning model
Loss function: feature map differences

What should be optimize against?
The deep learning model parameter?

Style transfer

Input

Content image



Style image



Synthesized image



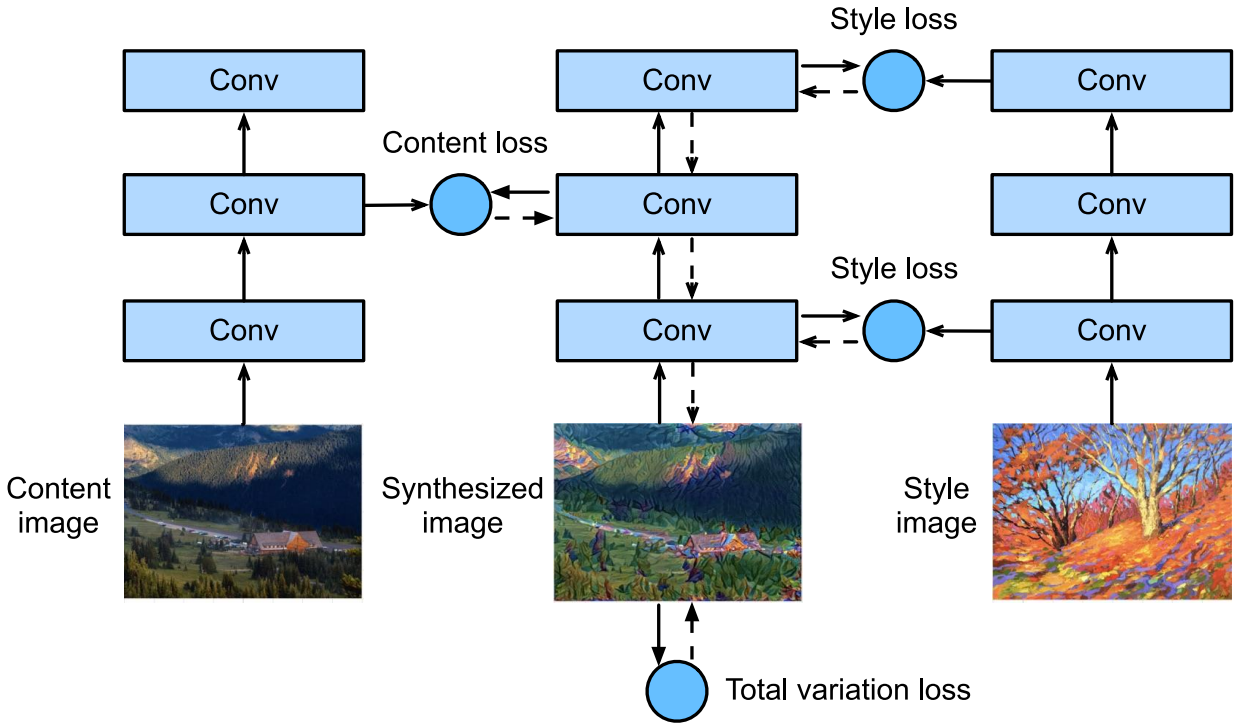
Output

Feature map in the deep learning model

Loss function: feature map differences

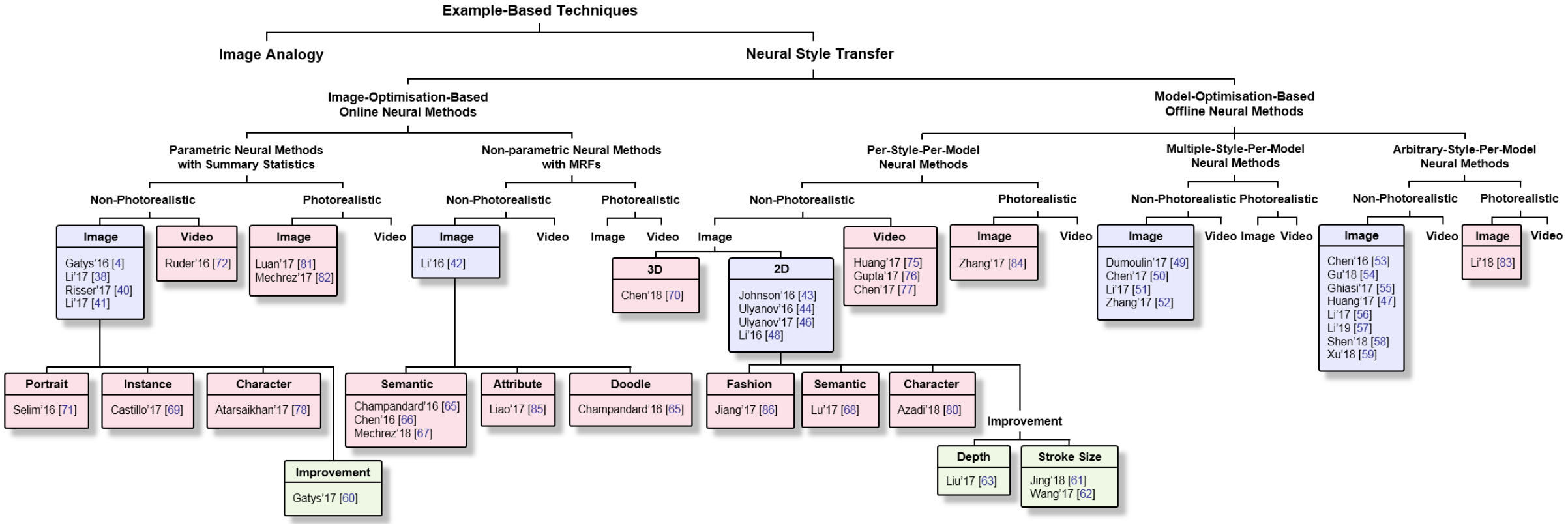
We optimize each pixel in the output image and fix the pretrained deep learning model

Style transfer



Model: a pretrained DL model
Parameters: every pixel in the output image
Loss function: content loss, style loss, and variation loss

Neural style transfer



Today's agenda



- About writing

Project proposal



- Due: Oct 15

- 1 page report (you can write as many as you want to), clarifying the following things
  - Title, Author(s)
  - (25%) What is the problem that you will be investigating? Explain the task thoroughly. Why is it interesting?
  - (25%) What is the data you will be using? Please include relevant characteristics such as the source of the data, the size of the dataset, and a sample of the data. Explain clearly which parts of the data will be used as well as potential obstacles.
  - (25%) What methods do you plan to experiment with? Please thoroughly describe them. If there are existing related implementations, will you use them and how? How do you plan to improve or modify such implementations? This can be subject to change, but you should have a general sense of how you plan to approach the problem you are working on.
  - (25%) How will you evaluate your results? Qualitatively, what kind of results do you expect (*e.g.,* plots or figures)? Quantitatively, what kind of analysis will you use to evaluate and/or compare your results (*e.g.*, what performance metrics or statistical tests)? What is your hypothesis regarding your results compared to baselines?

From lecture 1

Problem	Average
[Identifying the course project]	2.475
[Paper writing skill]	2.125
[Understand the mathematical background of the deep learning methods]	2.1
[Debugging the deep learning programs]	2.075
[Finding additional materials and resources to learn more about the concepts]	1.95
[Implementation the methods with programming]	1.825
[Understand the concept in deep learning]	1.775
[Presentation skill]	1.725

Why do you need to care about writing?



- News/information communication
- Report your progress to your boss
- Show your knowledge: HW/exams
- Store information
- …

Writing is important for your daily life, not just academia

What is the structure of an academic paper?



- Introduction
- Related works
- Problem definition/Methods
- Results
- Discussion and Conclusion

What is the structure of an Introduction?



- What is the topic? Why do we want to study it? Why do we want to propose computational methods to study it? How computational methods can contribute?

- What are the previous computational methods studying it?
  - Categorize the methods and summarize the key idea
  - Point out their advantages, and especially, their disadvantages (if they are already perfect, why do you propose a new method)
  - Summarize the common disadvantage of the existing methods
  - Summarize the difficulties of the topic

- Introduce your method
  - Summarize your idea and method
  - Introduce how your method resolve the aforementioned difficulties and disadvantages
  - What is the rationale behind your method to make it work
  - Summarize the results

- Summarize your key contributions
  - Why it should be published and why your targeting audience would find it interesting to read

Related works



- Summarize the related works and distinguish your method from them (novelty)
  - Machine learning methods
  - Other related computational methods
  - Categorize them
  - An extension of the introduction part to emphasize on your novelty

Methods



- Overview of your methods
- Problem definition
- Depends on what the novel parts of the methods are
  - The features
  - The model
  - Loss function
  - New learning strategy
  - …
- Other interesting ideas involved in your method
- Multiple subsections and each paragraph should be detailed and logical enough
  - What/Why/How
  - Avoid 2-line paragraphs, which means you do not explain things clearly enough

Structure the result part as an appealing story



- Overall cross-validation performance
  - Not enough, maybe overfitting
- Cross-dataset/species/time validation
  - Good, but do your method really work for the real-life problem?
- Experimental validation
  - Great, but why do your method work and how to explain the results?
- Ablation study and feature analysis
- …
- Eventually, we will prove our method is a novel, useful, and transparent method, without any doubt

Discussion and Conclusion



- Summarize your method and contribution
- Discuss why your method can work
- Discuss how useful your method would be
- Discuss how your method can potentially be further improved (a new direction or some perspective)
  - Why did not you implement them in this project?
- Conclude the paper

A few tips



- As non-native English speakers, we should emphasize the LOGIC of the paper
  - The transition between each sentence should be smooth, and the entire paper should be structured logically
  - What is the topic? Why do we want to study it? Why do we want to propose computational methods to study it? How computational methods can contribute?

- Read papers from your targeting conference and journal, knowing their favour
  - Nature-family journals care more about the problem and impact while conference usually look for novel technical details
  - The general guidance is the same while the details and style can be different

- The paper should be readable, at least for the targeting audience. I usually try to make the paper as clear as possible

A few tips



- Read your own paper
  - If you do not like your paper, why the reviewers would find it interesting
- Make appealing and professional figures
  - Check the figures from the nature family journals
  - Do not use the default setting when using PPT to make figure, too ugly
- Eliminate typos and grammar errors
  - Grammarly would not work all the time
  - Read your own paper
- Repeat the important points along the entire paper
  - Repeating is not a problem while not conveying the important idea is a big failure

A few tips



- Start writing as early as possible
  - After literature review, you can already start writing the first two paragraphs of the introduction part
  - Start writing is the most difficult thing
  - People would usually defer the writing to the last minute

- Keep revising your paper
  - You can write garbage in the first draft, which is total OK
  - But you need to keep reading, revising and rearranging your draft into the final paper

- Read and mimic the high-profile papers. Prepare for higher standard than your targeting venue
  - Nature papers
  - Conference high-quality/best papers

What do we mean by LOGIC?



- RNA binding proteins (RBPs) play essential roles in regulating the transcription, metabolism, and translation of cellular RNAs. *(Research area)*
- Determining RBP binding profiles in different conditions and elucidating their detailed regulatory mechanisms are critical for understanding their functions. *(Research problem)*
- However, given the sheer number of RBPs that account for close to 10% of the human proteome, establishing links between RBPs and their targets has been an enormous challenge. *(Specific research problem in this paper)*
- To address this question, many high-throughput technologies have been developed to profile and predict RBP binding. Assays such as systematic evolution of ligands by exponential selection (SELEX), RNAcompete, and RNA Bind-n-Seq can characterize the sequence preferences of RBPs in vitro, and methods like RNA immunoprecipitation (RIP) and UV crosslinking followed by immunoprecipitation (CLIP) and sequencing can identify RBP binding sites in vivo. *(Previous works)*

What do we mean by LOGIC?



- In addition to the methods of direct measuring, other approaches have been developed to model and predict RBP binding. Traditionally, position-weight-matrices have been used to describe RBP binding determinants and to predict RBP binding targets from RNA sequences. Machine learning methods that integrate different types of information also have been developed to more accurately characterize the binding pattern of RBPs. More recently, deep learning approaches have been successfully applied to model protein–RNA interactions and predict RBP binding sites. For example, DeepBind was developed to learn RBP binding preferences from RNAcompete data using a deep neural network. *(More previous works in another category)*

What do we mean by LOGIC?



- Although these learning methods successfully capture RBP binding preferences of primary sequence, their prediction accuracies under different physiological states are limited because the RNA sequence is independent of in vivo conditions. Over the years, several methods have been developed to include RNA structural features of RBP targets in their modeling, but these structures were based on computational prediction rather than in vivo analysis. Although RNA structure can be predicted from sequence with some accuracy, the predictions do not reflect the dynamic regulations by cellular trans-factors and usually show substantial differences from in vivo structures. Thus, in vivo RNA structure data are essential for accurate modeling and predictions of protein-RNA interactions in physiologically relevant contexts. *(Limitations of the previous research, motivation)*

What do we mean by LOGIC?



- Here, we bridge this knowledge gap by determining transcriptome-wide RNA secondary structures in multiple cell types. We then integrate this experimentally-derived structure information in the construction of a deep discriminative neural network Protein-RNA Interaction by Structure-informed Modeling using deep neural NETwork (PrismNet) that accurately models and predicts RBP targets in vivo. We apply PrismNet to predict how genomic variants affect RBP binding, especially in the context of human diseases. Specifically, we focused on single nucleotide variants that disrupt RNA structure (riboSNitches) and are often associated with human disease, and discovered that riboSNitches are enriched in dynamic, cell-type-specific RBP binding sites. *(Summary of the proposed method)*

Illustration examples

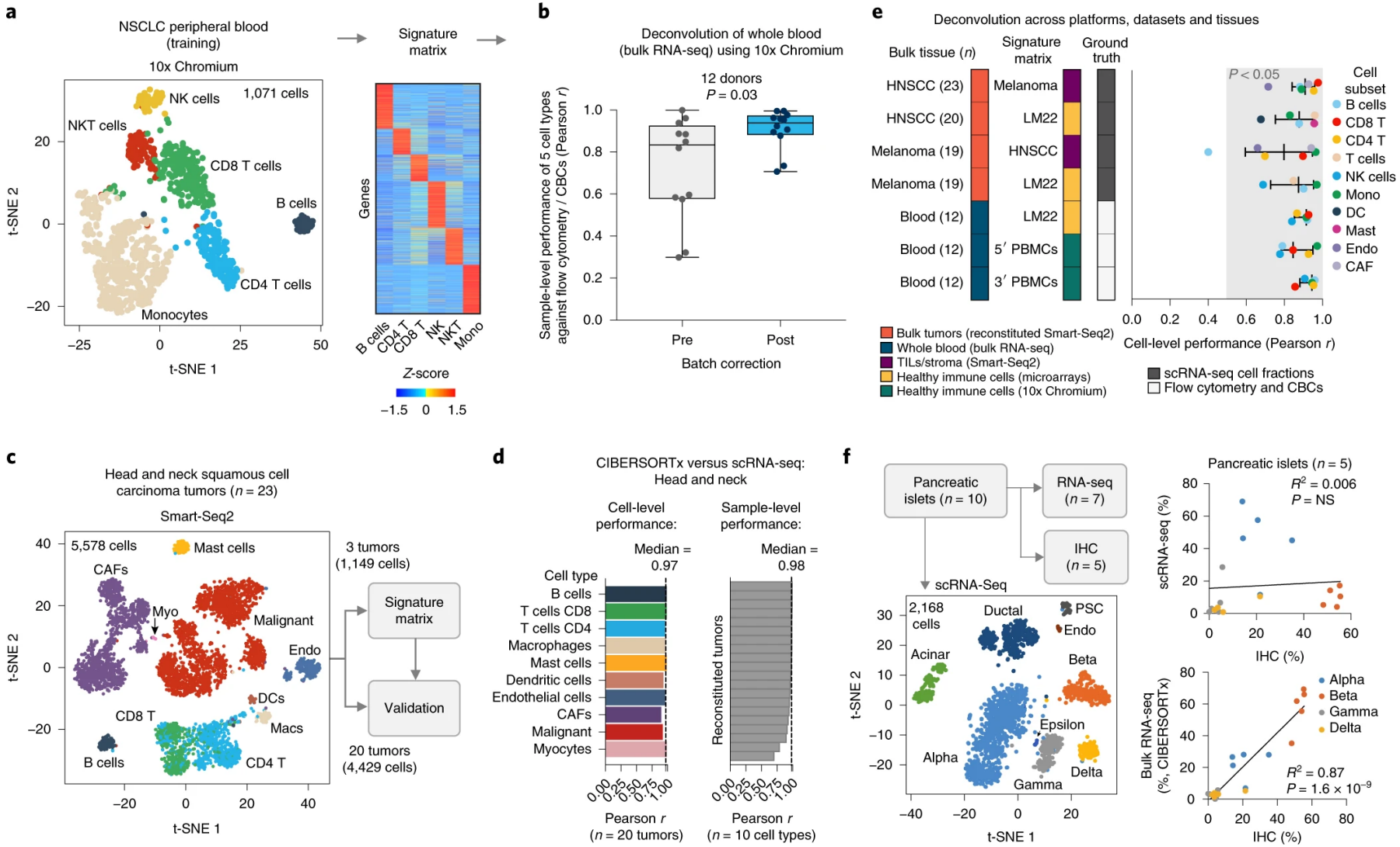
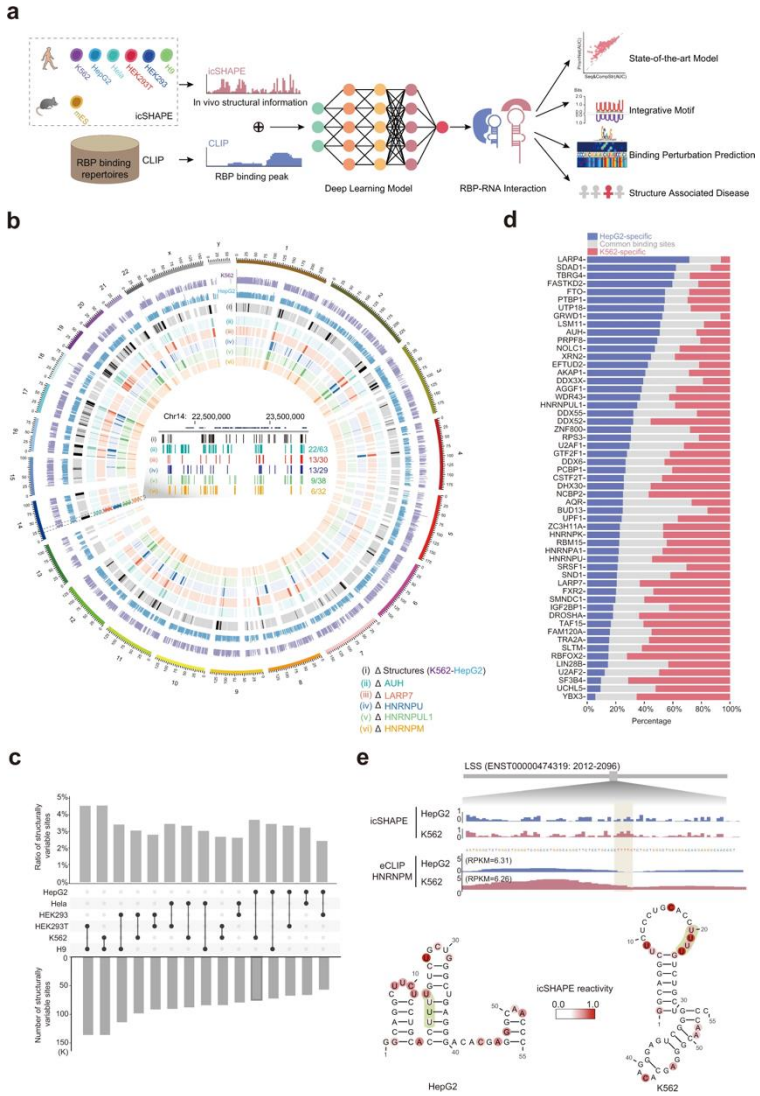
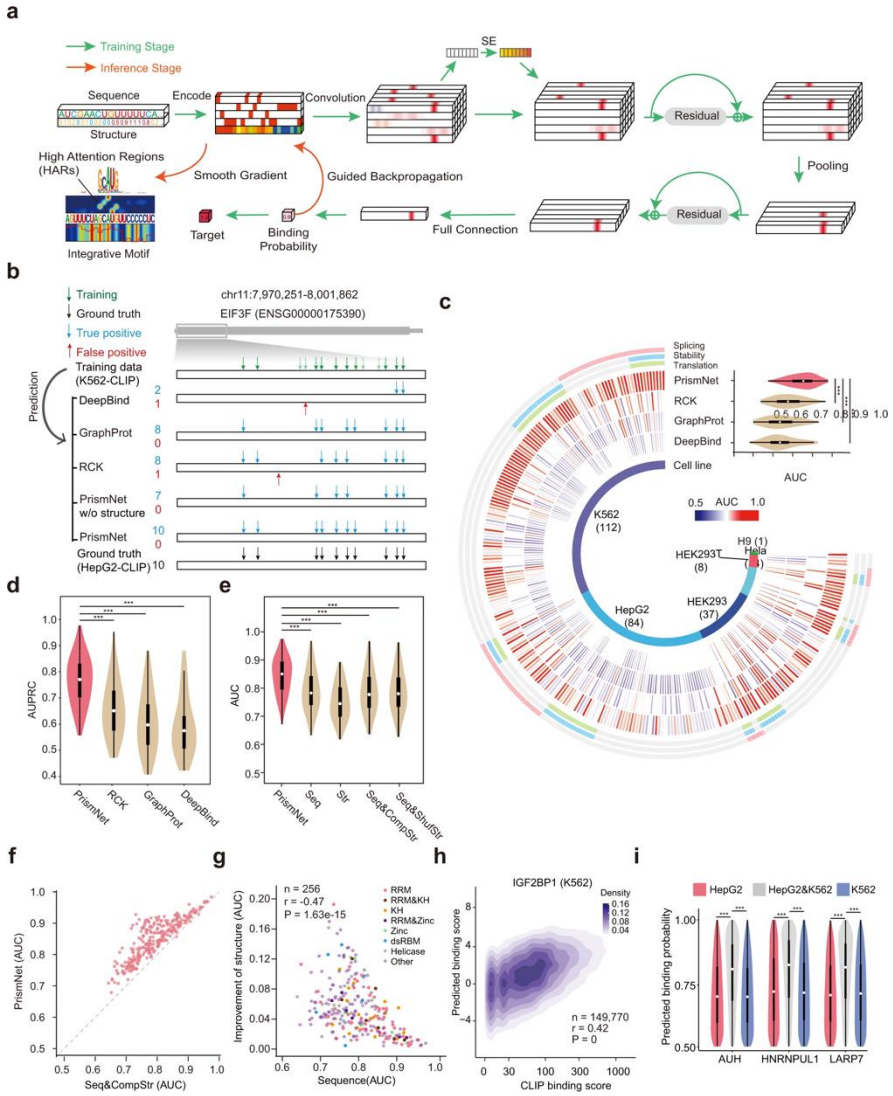


Illustration examples





Take home message



- Academic writing would benefit your career even if you decide to go to industry after graduation

- For the course report
  - You are not required to reach that level of quality
  - But you need to realize that you should never feel satisfied with your writing skills

Resource and advanced topics



- Dive into deep learning: Chapter 12, 13, 14

- Science Research Writing A Guide for Non-Native Speakers of English
- Data visualization
  - https://courses.cs.washington.edu/courses/cse512/21sp/

Post-lecture survey

- https://forms.gle/VwBH1qDyuRmGQcFN6

Next time



- Loss function and Hyper-parameters



Thank you!

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Department of Computer Science and Engineering (CSE)

The Chinese University of Hong Kong (CUHK)